Title: User Manual for Cyanobacteria Aggregated Manual Labels (CAML) Dataset

Summary: Continuous monitoring for cyanobacteria blooms in small, inland water bodies via *in-situ* sampling and analysis can be challenging not only due to the number and locations of water bodies to cover, but also due to the dynamic nature of algal growth and toxin production. Detection targets vary with cyanobacteria strains as well as physical, chemical, and biological factors. Ground monitoring also lacks consistency as sampling methods, frequency, and analytical techniques vary from region to region. However, remote sensing allows systematic data collection over a large area to identify regions with potential harmful algal growth. We introduce the Cyanobacteria Aggregated Manual Labels (CAML), a large dataset of *in-situ* cyanobacteria measurements for investigations of cyanobacteria detection and severity classification in inland water bodies across the United States. Relevant satellite imagery from publicly available endpoints are applicable to use when applying the CAML dataset to models. The dataset labels ground measurements of cyanobacteria cell counts at 23,570 points in U.S. inland water bodies over 2013 – 2021. Algorithms trained on this data could be used to estimate cyanobacteria cell counts in water bodies for timely water quality and public health interventions and to gain an understanding of environmental and anthropogenic factors associated with cyanobacteria incidence and proliferation. Data is provided in a comma-separated values (CSV) format.

Browse Image:

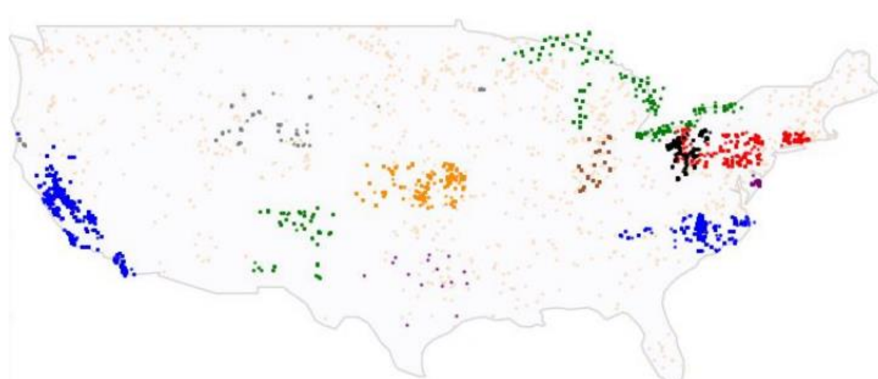

Image File Name: Cyanobacteria Aggregated Manual Labels CAML Dataset Study Area.png

Image Description: Distribution of Cyanobacteria Aggregated Manual Labels (CAML) Dataset across the United States. Each point represents a piece of data.

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DATASET OVERVIEW

Harmful algal blooms (HABs) are characterized by a rapid increase in the concentration of algal biomass or toxin production within a freshwater or marine system (Shen et al. 2012). Blooms are influenced by many factors including water temperatures, eutrophication, pollution, and changes in the food web. Different species of HABs can produce toxins that deplete oxygen supplies in water bodies and contaminate drinking water resources (Mehta and Palacios 2017). Early detection and regular monitoring for HABs can improve aquatic ecosystem health and water quality across the globe.

The Environmental Protection Agency (EPA) suggests four steps when analyzing smaller water bodies for cyanobacteria concentrations (EPA, 2019):

1. Assess the water body for the presence of HABs and focus on recreational waters for monitoring
2. Observe recreational water bodies for HABs throughout the recreational season
3. Monitor for cyanotoxins specifically for environmental and human health
4. Follow-up on any positive cyanotoxin monitoring

These steps require water quality managers to have consistent monitoring for recreational water bodies within their jurisdiction. *In-situ* measurements of cyanobacteria are gathered at affected water bodies and tested in laboratories for cell density and cyanotoxins (EPA, 2019). Only *in-situ* samples can determine the presence of cyanotoxins within a water body (EPA, 2019). This method of sampling is time consuming as it assumes that the water quality managers can routinely sample at multiple sites in a large number of water bodies, requiring personnel support, adequate sampling materials, and capacity for laboratory analysis. If impacted water bodies are on opposite ends of the state, water quality managers may have to prioritize or limit sampling trips to be able to effectively monitor all impacted areas. For example, California has 155,779 square miles of land across the state (USGS, 2010). Of that land, 2,833 square miles are inland water making up about 1.8% of the land of California (USGS, 2010). While the area of inland water bodies within the state seems low compared to the overall area, this is still a large amount of water to monitor continuously with unpredictable changes occurring throughout water bodies. Finding a more sustainable solution to support *in-situ* monitoring would be beneficial for water quality managers, specifically during recreational seasons.

Satellites can accurately measure reflected solar radiation in multiple spectral bands (Mehta and Palacios 2017) which allow for enhanced spectral, spatial, and temporal resolution of HABs (Mehta and Palacios 2017). Particularly, the concentration of Chlorophyll-a (Chl-a) and sea surface temperature (SST) are common HAB detection variables that can be estimated from satellite data (Mehta and Palacios 2017).

Several challenges remain for monitoring of HABs in inland and coastal water bodies. First, smaller water bodies are harder to observe because of their size with respect to the limited spatial, temporal, and spectral resolution of remote sensing technology (Kudela et al. 2015). Finer resolution and more frequent imaging is needed for better detection of smaller HABs

(Kudela et al. 2015). Secondly, algorithms to properly process data are needed, as presently they are limited in capacity for detecting smaller blooms (Kudela et al. 2015). Finally, there is no standard national or international system for monitoring HABs, resulting in inconsistent methods and targets of detection (Hill et al. 2020).

We introduce the Cyanobacteria Aggregated Manual Labels (CAML), a dataset for cyanobacteria classification. CAML covers the greater United States area and represents the largest dataset with *in-situ* measurements that we are aware of. We detail the design considerations behind CAML and the process by which the data were collected.

Related Publications: N/A

Related Datasets: N/A

Acknowledgments: This work was made possible through support provided by the National Aeronautics and Space Administration (NASA), United States Geological Survey (USGS), National Oceanic and Atmospheric Administration (NASA), Environmental Protection Agency (EPA), and DrivenData. We thank Tracy Fanara, John Haynes, Amy Kaminski, Keith Loftin, , Nima Pahlevan, Blake Schaeffer, Joel Scott, Ryon Stewart, Richard Stumpf, Michelle Tomlinson, and Katie Ward for their support in research and development for this dataset and subsequent model development. Thank you to Trista Brophy Cerquera for her support in the analysis of algorithms developed through the Tick Tick Bloom Competition (DrivenData, 2023). This work was funded under award number 80JSC022F0197.

Information for this work was provided by Bureau of Water Kansas Department of Health and Environment, California Environmental Data Exchange Network, Connecticut State Department of Public Health, Delaware National Resources, EPA Central Data Exchange, EPA National Aquatic Research Survey, EPA Water Quality Data Portal, Indiana State Department of Health, New Mexico Environment Department, North Carolina Division of Water Resources, North Carolina Department of Environmental Quality, Pennsylvania Department of Environmental Protection, Texas Commission on Environmental Quality, U.S. Army Corps of Engineers, University of Delaware's Citizen Monitoring Program, USEPA GLNPO’s Monitoring Program, and Wyoming Department of Environmental Quality.

DATASET CHARACTERISTICS

Spatial Coverage: The United States of America

Spatial Resolution: Not Applicable

Temporal Coverage: 2013-01-04 00:00:00 to 2021-12-29 23:59:59

Temporal Resolution: 10 day return (Suggested parameters)

Study Area: Bounding box estimates: 26.38943, -124.179; 48.97325, -67.6987

Global Change Master Directory (GCMD) Keywords (optional): Provide keyword(s) relevant to the dataset Locations, Earth Science field, and measurement Platforms and Instruments (available at https://gcmd.earthdata.nasa.gov/KeywordViewer/).

Cyanobacteria, blue-green algae, earth science, water quality, United States,

File Naming Convention: The dataset file was named for the content included.

File Descriptions: There is one file that includes all metadata and the data itself.

Data File Properties: All data are provided in a CSV format.

Data Details:

Each *in situ* data point in CAML represents the following information.

- uid (str): unique ID for each row
- data_provider (str): Name of the organization that provided the measurement
- region (str): Region of the U.S. Possible values are midwest, northeast, south, and west
- latitude (float): Latitude of the location where the sample was collected
- longitude (float): Longitude of the location where the sample was collected
- date (pd.datetime): Date when the sample was collected, in the format YYYY-MM-DD
- timestamp_gmt (pd.time): Time the sample was collected, in the format HH-MM. Sample time is not available for all data points. For samples without a recorded 'timestamp_gmt'

value, the ‘timestamp_gmt’ field was populated with value ‘00:01’. No actual samples were collected at 00:01 GMT.”

- density_cells_per_ml (float): Raw measurement of total cyanobacteria density in cells per mL
- severity (int): Severity level based on the cyanobacteria density. The density ranges for each severity level are:

severity level	**Density range (cells / mL)**
1	<20,000
2	20,000 - <100,000
3	100,000 - <1,000,000
4	1,000,000 - <10,000,000
5	≥10,000,000

- distance_to_water_m (float): Euclidean distance to the nearest water body in meters. This column can be used to identify sample locations that may be noisy or incorrect. For example, a distance of 0 indicates that the point is in a water body. Distances were calculated using ESA's World Cover map and Google Earth Engine.

Other Details: Dataset statistics provided by DrivenData.

Table 1. Cyanobacteria cell density samples provided by data providers.

Table 1. Cyanobacteria cell density samples provided by data providers	
Data provider	**Samples**
N.C. Division of Water Resources N.C. Department of Environmental Quality	10,902
California Environmental Data Exchange Network	5,645
Bureau of Water Kansas Department of Health and Environment	1,465

US Army Corps of Engineers	1,145
EPA Central Data Exchange	1,086
EPA National Aquatic Research Survey	894
Pennsylvania Department of Environmental Protection	836
Indiana State Department of Health	649
Connecticut State Department of Public Health	325
Delaware National Resources and the University of Delaware's Citizen Monitoring Program	221
New Mexico Environment Department	152
EPA Water Quality Data Portal	139
Wyoming Department of Environmental Quality	96
Texas Commission on Environmental Quality	15

Table 2. Cyanobacteria cell density samples ranked by severity level.

Table 2. Cyanobacteria cell density samples ranked by severity level	
Severity level	**Samples**
1	9,761
2	4,083
3	3,812

4	5,824
5	90

Table 3. Cyanobacteria cell density samples organized by year of collection.

Table 3. Cyanobacteria cell density samples organized by year of collection	
Year	**Samples**
2013	1,883
2014	2,106
2015	2,826
2016	3,053
2017	3,466
2018	2,717
2019	2,987
2020	2,252
2021	2,280

Table 4. Cyanobacteria cell density sample breakdown by distance to water.

Table 4. Cyanobacteria cell density

sample breakdown by distance to water	
Distance to water (m)	**Samples**
0	6,722
1 - 100	2,956
101 - 1,000	11,076
1,001 - 2,000	2,286
>2,000	529

Table 5. Cyanobacteria cell density sample breakdown by season.

Table 5. Cyanobacteria cell density sample breakdown by season		
Seasonal Distribution	*Months of Interest*	*Data Density*
Summer	June, July, and August	10,813
Spring	March, April, and May	5,045
Fall	September, October, and November	4,758
Winter	December, January, and February	2,954

Table 6. Percentage of cyanobacteria cell density samples with sample time available by data provider.

Table 6. Percentage of cyanobacteria cell density samples with sample time available by data provider

Data provider	**Percent with sample time**
California Environmental Data Exchange Network	100%
Connecticut State Department of Public Health	100%
Delaware National Resources and the University of Delaware's Citizen Monitoring Program	100%
Indiana State Department of Health	100%
Pennsylvania Department of Environmental Protection	100%
Texas Commission on Environmental Quality	100%
US Army Corps of Engineers	100%
Wyoming Department of Environmental Quality	99%
EPA Water Quality Data Portal	96%
New Mexico Environment Department	17%
Bureau of Water Kansas Department of Health and Environment	0%
EPA Central Data Exchange	0%
EPA National Aquatic Research Survey	0%
N.C. Division of Water Resources N.C. Department of Environmental Quality	0%

APPLICATION & DERIVATION

Existing Data Sources

Previous research has shown remote sensing to help identify algal blooms in water bodies, but this has been primarily focused on large water bodies such as oceans. Studies have shown that there are three forms of algorithms used when trying to understand pigments by satellites: analytic, semi-analytic, and derivative (Stumpf et al., 2016). Both analytic and semi-analytic methods require atmospheric corrections while derivative methods have shown that they can predict chlorophyll-a concentrations (Stumpf et al., 2016). What remains a challenge in creating these algorithms is the variability among *in-situ* samples and establishing what aspect, or indicator, of algae to study. Studies show that remote sensing can be used to create pixel based clusters in order to better understand algal bloom formation and behavior (Coeffer et al., 2021). Successful algorithms have been developed to understand the distribution of chlorophyll-a for inland water bodies within the United States, but not for cyanobacteria (Seegers et. al, 2021). We present a dataset that encompasses *in-situ* measurements of cyanobacteria cell density. This dataset is to support the generation of algorithms for the detection of cyanobacteria in smaller and inland water bodies. The CAML dataset creates a standardized dataset and unit of measurement for classifying cyanobacteria across multiple geographical systems.

When using ocean color satellites to understand algal blooms in water, the proper signal to noise ratio is necessary as well as the proper spectral bands. Sentinel 3 Ocean and Land Color Instrument (OLCI), for example, can detect chlorophyll and distinguish cyanobacteria, but has limited applicability in inland water bodies due to its coarse spatial resolution (300 meters by 300 meters) (Wolfe, 2023). Ocean-observing satellites have a greater spatial resolution and can only detect approximately 1-6% or 2,200 of the 200,000 lakes in the United States due to their coarse resolution (Clark et. al, 2017). So to understand water quality in smaller and more inland water bodies has proven to be difficult because more lakes need to be analyzed. A finer spatial resolution is necessary to monitor smaller water bodies. The CAML dataset provides a step forward in inland water quality monitoring.

In October of 2015, the Environmental Protection Agency (EPA), the National Aeronautics and Space Administration (NASA), the National Oceanic and Atmospheric Administration (NOAA), and the United States Geological Survey (USGS) created the CyAN Project (https://www.epa.gov/water-research/cyanobacteria-assessment-network-cyan); a joint-agency effort to support algal bloom monitoring within freshwater bodies throughout the United States (Schaeffer et. al, 2015). This project helps to promote further development of understanding algal blooms in freshwater ecosystems on a larger scale with public input (Schaeffer et. al, 2015).

Cyanobacteria Datasets

Water quality data are primarily available online through two sources: water quality managers and academic literature. Water quality managers across the US had a variety of algal bloom monitoring data including concentrations of chlorophyll-a, microcystin, and cyanobacteria cell density. In addition to inconsistency in the variables monitored by various state and local organizations, they also used different sampling methods, varied in reporting criteria and thresholds for their data repository, and in formats in which they stored data. The majority of data found through initial internet searches included chlorophyll-a and microcystin data. Some datasets only cataloged advisories or warnings about blooms, but contained no specific sampling information. Building the CAML dataset required in-depth research to locate cyanobacteria cell density measurements from *in-situ* monitoring.

QUALITY ASSESSMENT

Overview of the CAML Dataset

In-situ cyanobacteria measurements are stored and provided to the public in ad hoc, non-standardized formats and download mechanisms. Different classifications and identifiers for cyanobacteria were measured using different sampling methods, labeled with different units, formatted differently, and became difficult to gather. The CAML dataset represents measurements from 14 state and local water quality monitoring organizations (Table 6), and standardizes the use of cyanobacteria cell density in cells/mL for monitoring water quality. Dataset validation ensured that cyanobacteria measurements were represented in consistent units of cells/mL, that sampling information verified that one measurement was conducted per location, metadata included latitude and longitude coordinates as well as date of sampling, and corresponding satellite imagery were gathered. Cyanobacteria cell density measurements were labeled as 1 (low) - 5 (high) based on a custom severity scale described below (Table 7). The CAML dataset consists of 23,570 pieces of cyanobacteria cell density data (Figure 1).

83% of the CAML dataset (19,581 points) was collected between the years 2015-2021 (Table 3). This is likely due to collection records being more easily accessible in a digital format from 2015 onwards.

DATA ACQUISITION, MATERIALS & METHODS

Working with Cyanobacteria Cell Density Data

Algal blooms can be identified by chlorophyll-a, microcystin, or cyanobacteria cell density (EPA, 2019). Chlorophyll-a is a common measurement associated with identifying algal blooms in water, but typically is used to measure the amount of phytoplankton present within a water body ([2] EPA, 2022). Microcystin is a measurement more commonly used to understand the amount of toxins present within a water body, however, its presence cannot be inferred from satellite imagery (EPA, 2023). *In-situ* measurements are the only current accurate way to detect microcystin in water bodies (EPA, 2023).

Cyanobacteria has the potential to be understood as a measurement for algal growth on a larger scale compared to chlorophyll-a (Kislik et. al, 2022). While chlorophyll-a is easier to identify using Sentinel 2 imagery, this detection target has limitations on a large scale due to available data, spatial resolution in smaller water bodies, and band wavelengths for understanding toxins (Kislik et. al, 2022). Cyanobacteria cell density measurements were selected as they represent the amount of cyanobacteria cells present based on a water body (EPA, 2019). This direct measurement is accurate in understanding the presence and amount of algae in water bodies (EPA, 2019). It is important to also recognize that further, likely *in-situ*, data collection and analyses are necessary to classify the specific algal species and to determine potential for human, animal, and environmental harm from any released toxins (EPA, 2019).

Data Range of 2013-2021

The time period of 2013-2021 was selected for developing the CAML dataset. This time period gave 8 years of analysis, allowing a long duration to analyze changes and frequencies of algal blooms across the United States. According to the EPA, 2012-2021 was the warmest decade recorded with thermometer readings ([1] EPA, 2022; NASA/GISS, 2022). The EPA also reported that the years 2016 and 2020 set records for the warmest years recorded with thermometers readings ([1] EPA, 2022). The United States appears to be warming faster than the global average warming rate with an average increase of 0.17 degrees F ([1] EPA, 2022). The CAML dataset design team focused on analyzing algal blooms within the past decade due to this global warming as higher temperatures have an influence on the rate of algal blooms ([1] EPA, 2022; Mehta and Palacios 2017).

Establishing a Severity Scale Using Cells/mL

As ground measurement data were gathered from end-users and water quality managers, a variety of sampling methods and units were provided to our team. To ensure standardization among *in-situ* measurements for cyanobacteria cell density, the units of cells/mL were chosen. Measuring cyanobacteria cell density in cells/mL is the standard for the World Health Organization (WHO) (WHO, 2003). The unit of cells/mL is also the most common and transferable among laboratory and sampling methods (WHO, 2003). Setting cells/mL as the standard unit of measurement for cyanobacteria cell density allowed the team to organize *in-situ* measurements effectively without conversions from proxy indicators, average among *in-situ* sampling methods, and establish a common unit for understanding the algal blooms. The CAML dataset design team used these measurements to create an algal bloom severity ranking adapted from WHO standards that translated cells/mL data into 5 classifications ranging from “low severity” to “very high severity”.

The World Health Organization (WHO) uses a scale of severity of low, medium, and high based on cell concentrations in cells/mL (WHO, 2003). We convened seven harmful algal bloom scientists from across the United States to discuss quantization points in the WHO’s cyanobacteria severity scale. Similar classifications utilized across the United States by local water authorities and thresholds selected for decision making (issuance of advisories, closing of water access, instigation of water treatment actions) were incorporated to create the cyanobacteria severity index used in CAML (Table 7).

Dataset Schema

Each *in situ* data point in CAML represents the following information.

- uid (str): unique ID for each row
- data_provider (str): Name of the organization that provided the measurement
- region (str): Region of the U.S. Possible values are midwest, northeast, south, and west
- latitude (float): Latitude of the location where the sample was collected
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severity level	**Density range (cells / mL)**
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2	20,000 - <100,000

3	100,000 - <1,000,000
4	1,000,000 - <10,000,000
5	≥10,000,000

- distance_to_water_m (float): Euclidean distance to the nearest water body in meters. This column can be used to identify sample locations that may be noisy or incorrect. For example, a distance of 0 indicates that the point is in a water body. Distances were calculated using ESA's World Cover map and Google Earth Engine.

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APPENDIX

Figure 1. Distribution of Cyanobacteria Aggregated Manual Labels (CAML) Dataset across the United States. Each point represents a piece of data.

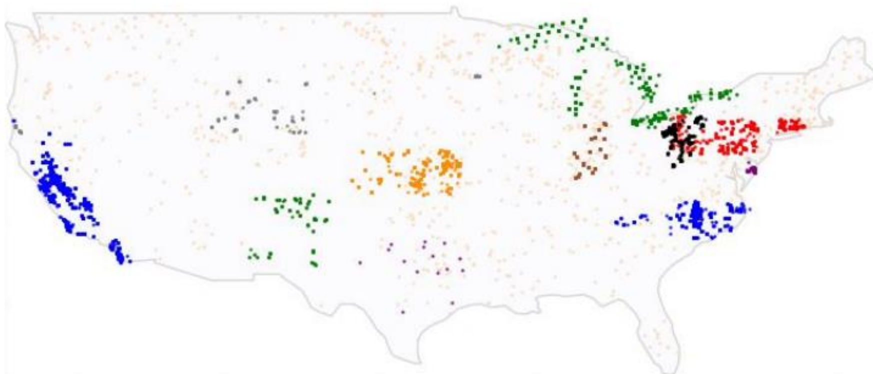

Table 7. Public Agencies, Organizations, and Research Groups that Provided Cyanobacteria Cell Density Data to Create the Cyanobacteria Aggregated Manual Labels (CAML) Dataset

Table 7. Public Agencies, Organizations, and Research Groups that Provided Cyanobacteria Cell Density Data to Create the Cyanobacteria Aggregated Manual Labels (CAML) Dataset				
**Organizations whose data were utilized for the CAML Dataset				
Upper Mississippi River Basin Association (UMRBA)	**US Army Corps of Engineers	Alaska Department of Environmental Conservation	Utah Department of Environmental Quality	**Pennsylvania Department of Environmental Protection
Georgia Environmental Protection Division - Watershed Protection Branch	Vermont Department of Health - Cyanobacteria (Blue-Green Algae)	Arizona Department of Environmental Quality	Rhode Island Department of Environmental Management - Office of Water Resources	Virginia Department of Health - Shellfish Safety

South Carolina Department of Health and Environmental Control - Harmful Algal Blooms (HABs)	**Indiana State Department of Health	Iowa Department of Natural Resources - State Park Beach Monitoring	West Virginia Department of Environmental Protection - Algae	**California Environmental Data Exchange Network (CEDEN) Advanced Tool
**Kansas Department of Health and Environment	Wisconsin Department of Natural Resources - Blue-Green Algae	Center for Disease Control (CDC) - Harmful Algal Blooms (HABs)	Louisiana Department of Environmental Quality	New Jersey Department of Environmental Protection - Division of Water Monitoring and Standards - Bureau of Freshwater and Biological Monitoring
**Wyoming Department of Environmental Quality	New Jersey Department of Environmental Protection - Division of Water Monitoring and Standards	Tennessee Department of Environment and Conservation	**New Mexico Environment Department	**Texas Commission on Environmental Quality
Maine Bureau of Water Quality - Division of Environmental Assessment	**Connecticut State Department of Public Health - Blue-Green Algae Blooms	New York State Department of Environmental Protection	**United States Environmental Protection Agency - Water Quality Data Portal	**Delaware Department of Natural Resources and Environmental Control
**North Carolina Department of Environmental Quality - Algal	United States Geological Survey (USGS) - Harmful Algal	North Dakota Department of Environmental Quality	**Environment al Protection Agency (EPA) National	**University of Delaware - Citizen Monitoring

Blooms	Bloom (HAB) Science in Texas		Aquatic Research Survey	Program
**Environmental Protection Agency (EPA) Central Data Exchange	Michigan Department of Environment, Great Lakes, and Energy - Harmful Algal Blooms	Ohio Environmental Protection Agency (EPA)		

Table 8. Severity levels of cyanobacteria cell density data using cells/mL. These severity levels were established based on the World Health Organization's (WHO) guidelines for monitoring algal blooms, and guidance from subject matter experts.

Table 8. Established Severity Levels of Cyanobacteria Cell Density Using Cells/mL		
Severity Level	*Classification*	*Cell Density (Cells/mL)*
1	Low	<20,000
2	Low	20,000 - <100,000
3	Middle	100,000 - <1,000,000
4	High	1,000,000 - <10,000,000
5	High	≥10,000,000